Chapter 31

Pharmacogenomics and Precision Medicine

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Learning Objectives

Objective 31.1 Define terms and explain basic concepts in pharmacogenetics/pharmacogenomics.
Objective 31.2 Explain the relationship of pharmacogenomics with pharmacokinetics and pharmacodynamics.
Objective 31.3 Describe pharmacist competencies in genomics/pharmacogenomics and their roles in precision medicine.

OBJECTIVE 31.1. DEFINE TERMS AND EXPLAIN BASIC CONCEPTS IN PHARMACOGENETICS/PHARMACOGENOMICS

Pharmacogenomics, as a component of overall genomics and precision medicine, requires the use of terms that are new to many pharmacists and student pharmacists. Table 31.1 presents the terms as used throughout this chapter. Although the terms are not entirely standard across all disciplines, the terms presented here are commonly used.

Optimization of patient care related to drug therapy includes providing a given patient with efficacious treatment while minimizing or avoiding the risk of adverse drug reactions/events.[1] The advent of pharmacogenomics as a component of data in clinical decision-making is facilitating optimal therapy from the start, avoiding "trial and error" therapeutics.[2]

The concept of genetic-related adverse drug reactions was discussed in 1957, and the term "pharmacogenetics" was applied in 1959 as researchers at the time were starting to investigate drug−gene interactions.[3,4] The broader term "pharmacogenomics" was born out of the recognition that likely many genes are involved in an individual's response to medication.[5] Rapid expansion of pharmacogenomic research, as a component of overall genomic research, occurred as the Human Genome Project (HGP), initiated in 1990, progressed to formal completion in 2003.[6] The HGP was intended to improve the basic science research infrastructure related to human genetics, have the human DNA sequence serve as the connection of human biology and the biology of model organisms, and to promote ongoing advancement in analytical biochemistry.[7] The advancement in analytical technology, which has continued post-HGP, has led to efficient and cost-effective approaches to the genotyping and sequencing of human DNA (Fig. 31.1).[8] The terms pharmacogenetics and pharmacogenomics have the abbreviations PGt and PGx, respectively.[9] PGx will be utilized as the common term in this chapter, except where explicitly cited.

Relative to PGx, the interactions of interest affect drug effectiveness as well as adverse drug reactions, with pharmacogenes involving drug metabolism, transport, interaction with receptors, enzymes, and/or alteration of biochemical signaling pathways.[10] When considering the influence of a drug−gene(s) interaction, it helps to recognize the consequence related to pharmacokinetics (PK) and/or pharmacodynamics (PD). As the majority of currently identified drug−gene interactions are related to drug metabolism, mainly considering the cytochrome P450 drug-metabolizing enzymes, it is important to understand what information is provided when PGx testing is undertaken.[11]

† Deceased. This chapter is dedicated to Dr. Brian S. Decker.

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TABLE 31.1 Terms and Definitions Utilized in Genetics/Genomics/Pharmacogenomics

Term	Definition
Allele(s)	Versions of the same gene inherited from each parent.
Biomarker (genomic)	Measurable indicators of normal biologic processes, pathogenic processes, and/or response to therapeutic or other interventions.
Gene	Regions of the genome (DNA) that contain the instructions to make proteins.
Genome	The entire DNA of an organism.
Genotype	The specific pair of alleles inherited at a locus on a given gene.
Heterozygous	Having two different alleles for the same trait.
Homozygous	Having identical alleles for the same trait.
Indel	Inserted or deleted DNA either as single nucleotides or spanning regions of DNA involving many nucleotides.
Intermediate metabolizer (IM)	Enzyme activity which is less than normal metabolizer activity and is more than poor metabolizer activity.
Monogenic trait	Features derived from a single gene.
Mutation	An alteration in DNA sequence between individuals.
Normal metabolizer (NM)	Full enzyme activity
Nucleoside/nucleotide	One of the structural chemicals of DNA, including adenine (A), cytosine (C), guanine (G), and thymine (T), and of RNA, including adenine (A), cytosine (C), guanine (G), and uracil (U). Nucleoside linked to a phosphate group.
Precision medicine	Utilization of patient-specific information and biomarkers to make more informed choices regarding the optimal therapeutic treatment regimen for a given patient.
Pharmacodynamics (PD)	The relationship between drug exposure and response.
Pharmacogenetics (PGt)	A single gene involved in response to a drug. Pharmacogenomics will be used as the preferred term, which encompasses pharmacogenetics.
Pharmacogenomics (PGx)	The study of many genes, and potentially the entire genome, involved in response to a drug.
Pharmacokinetics (PK)	The relationship of time and drug absorption, distribution, metabolism, and excretion.
Phenotype	An individual's expression of a physical trait or physiologic function due to genetic constitution and environmental and other factors.
Polymorphism	A mutation in DNA in a given population that may be observed at greater than 1% frequency.
Poor metabolizer (PM)	Enzyme activity which is less than IM activity. No too little enzyme activity.
Rapid metabolizer (RM)	Enzyme activity which is more than NM activity and is less than ultrarapid metabolizer activity.
Reference sequence number (refSNP, rs#, rs)	A unique and consistent identifier of a given single nucleotide polymorphism.
Single nucleotide polymorphism (SNP)	A variant DNA sequence in which a single nucleotide has been replaced by another base.
Theragnostics	Treatment approaches which combine therapeutics and diagnostics.
Ultrarapid metabolizer (UM)	Enzyme activity which is more than RM activity.

Genetic Variations

The nucleotide bases adenine (A), cytosine (C), guanine (G), and thymine (T) in ordered sequence comprise genes.[10] There are approximately 20,000 genes in the human genome, with the total sequence of DNA base pairs of the "double helix" being 3.2 billion. Variations in genes can be a consequence of a change in the sequence of the nucleotides with as little as

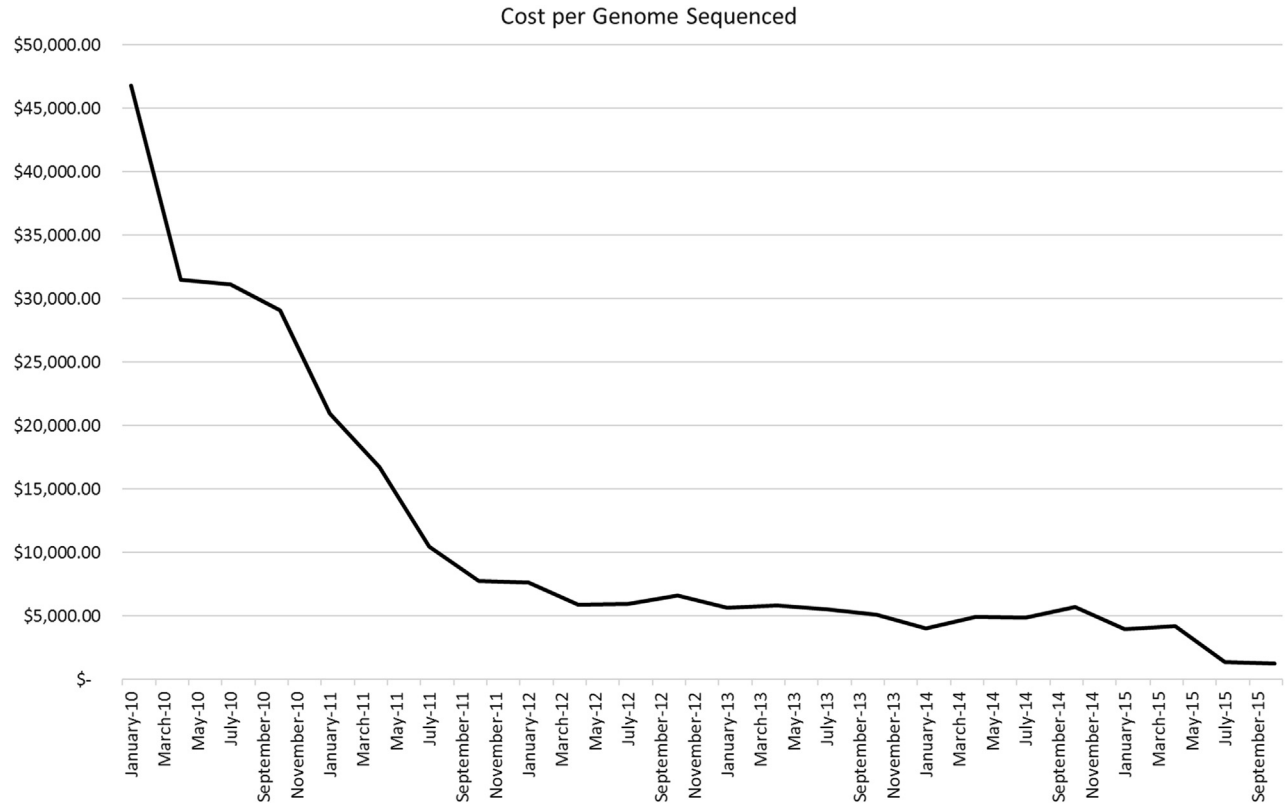


FIGURE 31.1 Decreasing costs of genetic testing either whole genome sequencing (shown above) or genotyping are leading to more economical applications of pharmacogenomics in the clinical setting.[8]

one base change resulting in altered gene product function/activity.[10] These single base changes, called **single nucleotide polymorphisms** (SNPs, "snips"), are the most common form of DNA variation.[10] Other forms of DNA variation include insertions and deletions ("indels"), which result in "shifting" of bases and altered sequences of amino acids and variant gene products.[10] Other DNA changes can also occur, such as the occurrence of gene duplication, where a number of copies of a given normal function gene (i.e., copy number variants (CNVs)) result in increased function. Broader discussion of genetic variation is beyond the scope of this chapter.

As SNPs are the most common form of DNA variation, defining the exact location of a given SNP in the DNA is noted by a reference sequence number or "rs" number.[12] An rs number is a consistent reference to a SNP that can be referred to across publications so the reader knows exactly what change is being discussed. The US National Center for Biotechnology Information houses an archived database of SNPs.[12] The database, dbSNP, is available for query and is the most comprehensive SNP database. As an example, the database can be searched for SNPs related to the gene *CYP2C19*. One such SNP is rs4244285, also known as c.681G>A, among other names, and defines the single change of A replacing G at position 681 of the coding region of the *CYP2C19* gene.[13] This SNP results in a no-function gene product CYP2C19 enzyme.[13] Examples of other SNPs and related rs numbers can be found below. By convention, a gene is noted in the text by italicized font (e.g., *CYP2C19*), whereas the gene product is not italicized (e.g., CYP2C19).

Nomenclature

In PGx relative to pharmacokinetic pharmacogenes, a "star (*)" nomenclature has been adopted to denote specific DNA changes.[14] For instance, rs4244285 is noted as *CYP2C19*2* in the star nomenclature. An international working group has recommended moving away from the star nomenclature, with specific recommendations.[15] However, PGx databases, such as the Pharmacogenomics Knowledgebase (PharmGKB) and guidelines from organizations (e.g., the Clinical Pharmacogenetics Implementation Consortium [CPIC] and the Dutch Pharmacogenetics Working Group [DPWG]), utilize the star nomenclature to a great extent.[16,17] Therefore, it is important to understand this nomenclature as PGx testing results, related to PK, are typically translated using the star nomenclature (e.g., A at rs4244285 is reported as *CYP2C19*2*).

To help clinicians avoid the pitfalls of nomenclature differences, standard metabolism phenotypes related to drug-metabolizing genes have been defined.[18] There are *five* defined metabolizer phenotypes including, from greatest metabolism to least metabolism,

- ultrarapid metabolizer (UM),
- rapid metabolizer (RM),
- normal metabolizer (NM),
- intermediate metabolizer (IM), and
- poor metabolizer (PM).

These phenotypes have been related to clinical application of PGx through guidelines, such as those from CPIC and the DPWG.[17,19] The metabolic phenotypes are related to an individual's genetics, being the basis for the inherent differences in maintenance dose requirements.[20] Other terms have been adopted for drug transporter genes and genes related to PD.[18] Individual rs numbers and metabolic phenotypes are referred to in the examples presented throughout this chapter.

Pharmacogenomic Testing in Pharmacotherapy

In the 2000s and 2010s, PGx testing costs have decreased dramatically and Illumina Scientific, a manufacturer of genotyping and sequencing instrumentation, has stated that the cost of sequencing an individual's whole genome (WGS) will drop to approximately $100 (USD).[21] To put this in perspective, the cost of WGS in 2001 was over $95 million (USD).[9] Even with lower technology costs, broad application of PGx in the clinical is still a major undertaking and there are many obstacles for transition to full implementation.[22–27]

Pharmacogenetic testing became prevalent with the discovery of monogenic (single) drug–gene interactions. With monogenic drug–gene interactions, clinicians order a pharmacogenetics test on an as needed reactionary basis; a clinician may order a test when effectiveness or an adverse drug reaction is of concern.[28] With the advancement of technology, and reduced costs, pharmacogenomic testing is more plausible in the clinic. Patients can now have numerous genes assayed at the same time, leading toward a preemptive approach that could guide clinicians in the care of individual patients over the remainder of their lifetime.[28–30] The preemptive approach could include genotyping/sequencing individuals when they are young (even at birth) and these results could be stored securely and queried throughout the patient's lifetime.[28–30] This preemptive approach would only include germline (inherited) DNA mutations, and any mutations occurring somatically (not inherited) may need to be tested later in life depending on the disease state (e.g., cancer).[11]

OBJECTIVE 31.2. EXPLAIN THE RELATIONSHIP OF PHARMACOGENOMICS WITH PHARMACOKINETICS AND PHARMACODYNAMICS

Many factors should be considered while clinically addressing a patient's drug therapy, with PGx being one of them.[31] PGx is important to consider as DNA variants can influence many physiologic-related processes, such as drug metabolism, and drug transport, resulting in ineffectiveness and/or adverse drug reactions. PK is the time course of drug absorption, distribution, metabolism, and excretion (ADME), i.e., the movement of a drug into and out of the body.[32] PGx can influence PK by altering the ADME of a given drug.[10] Therapeutic drugs are most often administered by the per-oral route and need to be absorbed and undergo other processes (e.g., distribution) to produce their intended effect. Orally absorbed drugs are characterized by their bioavailability, with a drug being bioavailable when it moves from the gastrointestinal lumen to systemic circulation having escaped gastrointestinal wall and "first-pass" metabolism by the liver.[10,32] For some drugs, active transport is required to facilitate movement across the enterocytes of the small intestine to bring drug to portal vein blood flow to the liver.[10] Both influx (taking drug up into intestinal epithelial cells) and efflux transporters (moving drug back to gastrointestinal lumen) can be altered by genetics, affecting the amount of drug that ultimately reaches systemic circulation.[33] Although a large majority of drugs cross the gastrointestinal wall via passive diffusion, influx and efflux transporters also influence drug movement (absorption) from the gastrointestinal lumen to portal blood flow.[10,33]

Additionally, the enterocytes of the gastrointestinal lumen and hepatocytes in the liver contain drug-metabolizing enzymes that can metabolize a drug before it reaches the systemic circulation.[10] Here, genetics can influence (i.e., reduce or increase metabolism) at these sites, affecting the amount of drug that reaches the systemic circulation. A number of drug-metabolizing enzymes in the gastrointestinal wall and liver are members of the cytochrome P450 superfamily and include CYP3A4/CYP3A5, CYP2C9, CYP2C19, and CYP2D6, among others.[34,35]

With the administration of a drug, via per-oral or any other route, numerous processes are set in motion. As discussed, on per-oral administration drug absorption leading to systemic exposure is required to elicit the desired effect. As drug is

distributed to various tissues of the body, it is also undergoing processes which result in a drug being "removed" from the body.[10,32] Many drugs require metabolic conversion to be eliminated from the body. Here, typically, a given drug is modified chemically to be more water soluble, thus promoting urinary excretion of drug metabolites. There are two types of drug metabolism: Phase I metabolism, which refers to chemical reactions such as oxidation, reduction, or hydrolysis, and Phase II metabolism, which refers to conjugation reactions such as glucuronidation, sulfation, acetylation, and methylation. Extensive research has revolved around Phase I-metabolizing enzymes such as the cytochrome P450 enzymes, but both phases of metabolism can be altered by genetics.[36] PGx can influence the metabolism of drugs by increasing the metabolic rate, which in turn clears the drug from the body faster (RM or UM). There can also be a decreased metabolic rate of drugs causing greater exposure as drug accumulation occurs (IM or PM). Typically, the majority of the general population are "normal" metabolizers of most drugs; however, genetic-related alteration in the metabolism of some drugs can result in inefficacy or serious adverse reactions (ADR). For instance, a given active drug may never be efficacious in a patient who is an UM, whereas the given active drug may result in an ADR in a PM.[37] Conversely, the administration of a prodrug to an UM may result in toxicity, whereas therapeutic failure may occur in a PM.[37] Other drug elimination processes include drugs or its metabolites being excreted from the body in the urine, bile, etc.[33] Influx and efflux transporters related to drug distribution and/or excretion, as noted above, can be influenced by genetics.

The clinical utility of PGx lies with how it influences drug response, either through PK or the PD of the drug. PD is the relationship between drugs and the mechanisms of their actions or the pharmacological response/effect in a given patient.[38] The variability in PD is usually greater than the PK variability and defining the drug response phenotype has become one of the most difficult components of PGx.[11]

The PD effects involve drug targets such as receptors, enzymes (nondrug metabolizing), and certain transporters (e.g., serotonin reuptake).[10] Drug targets are proteins and therefore can have altered function due to underlying DNA variation, such as SNPs. These polymorphisms can lead to decreased binding ability, a decrease in affinity for the drug receptor, and a decrease in the efficacy of the drug.[38] To add to the complexity, multiple drugs can act as either agonists or antagonists and enhance or inhibit the ability of the drug to elicit its effect(s).

Overall, drug response is likely due to multiple genetic factors that influence PK and/or PD. The influences of genetics on drug effects, related to inefficacy and/or toxicity, have now been noted in more than 140 package labels of drugs approved by the USFDA.[39] Below, specific examples of drug–gene interactions involving PK and/or PD are presented.

Example Drug–Gene Interactions

The following drug–gene pairs represent "model" interactions that are illustrative and can be used to understand similar pharmacogene effects across many drug–gene pairs.

Drug-Metabolizing Enzyme Pharmacogenes

CYP2C19-clopidogrel—For the specific indication of acute coronary syndrome postcoronary artery stent placement, clopidogrel is indicated as a component of antiplatelet therapy.[40] As a prodrug, clopidogrel needs to be converted into an active metabolite to elicit its antiplatelet effect. This conversion is facilitated by CYP2C19, the protein enzyme product of *CYP2C19*.[40–42]

More than 30 *CYP2C19* variants produce enzymes with different activities.[41] Table 31.2 provides example *CYP2C19* variants and related enzyme product activities.

The combination of alleles, called diplotypes, in Table 31.3 represented using the star nomenclature, leads to the assignment of metabolic phenotype designations. The CPIC undertook a survey process to identify standard terms for

TABLE 31.2 Example *CYP2C19* Variants and Resulting CYP2C19 Activity

CYP2C19 Variant (Base Change)/rs Number	Star Nomenclature Allele Designation	Enzyme Product Activity
c.681G>A/rs4244285	*2	No activity
c.636G>A/rs4986893	*3	No activity
c.-806C>T/rs12248560	*17	Increased activity

TABLE 31.3 CYP2C19 Diplotypes, Related Phenotypes, and Potential Therapeutic Consequences of Clopidogrel Use in ACS With PCI

CYP2C19 Diplotype (Star "*" Nomenclature)	Metabolic Phenotype	Clopidogrel Therapeutic Consequence in ACS With PCI of Stent Placement
*1/*1	Normal metabolizer (NM)	Clopidogrel use indicated
*1/*2	Intermediate metabolizer (IM)	Alternative to clopidogrel indicated
*2/*3	Poor metabolizer (PM)	Alternative to clopidogrel indicated
*1/*17	Rapid metabolizer (RM)	Clopidogrel use indicated
*17/*17	Ultrarapid metabolizer (UM)	Clopidogrel use indicated

consistency in reporting phenotypes.[18] Table 31.3 presents the various metabolizer phenotype examples relative to *CYP2C19* diplotypes. Considering clopidogrel in the setting of ACS and PCI with stent placement, the CYP2C19 phenotype has therapeutic implications (Table 31.3).

The resultant therapeutic consequence is a result of altered PK where the metabolism/conversion of the prodrug clopidogrel into its active metabolites is decreased.[43] The antiplatelet activity is altered, and patients with an IM or PM status are at increased risk of adverse cardiovascular events, including stent thrombosis and death.[42] The USFDA placed a boxed warning in the clopidogrel package labeling describing the reduced efficacy of the drug in PMs. The statement includes information about testing patients to identify their metabolic phenotype.[43]

CYP2D6-codeine—Tragic adverse drug reactions in children related to codeine use have been tied directly to PGx.[44] A number of deaths and serious ADRs to codeine in children have been attributed to individuals with an UM phenotype.[45,46]

Codeine is the prodrug of the active analgesic morphine. The conversion of codeine into morphine is mediated by CYP2D6, with UMs converting more codeine into morphine following a given dose. As there are more than 110 *CYP2D6* alleles resulting in more of a continuum of drug metabolism variation from the PM through UM phenotypes, an "activity score" system has been established to assign metabolism phenotypes.[41,47] Activity scores of each allele are summed to provide the overall activity score. Alleles with an activity score greater than 1, as is seen with gene duplication, are considered to have increased function; those with an activity score of 1 are normal function; those with an activity score of 0.5 are decreased function alleles; and those with an activity score of 0 are no-function alleles.[47] Table 31.4 provides examples of *CYP2D6* alleles *CYP2D6* diplotypes, the related overall activity score and the metabolic phenotype. It should be noted that other approaches to defining the activity of CYP2D6 are being considered in an effort to better qualify genetic influence.

The frequency of alleles in a given population varies considerably. Table 31.5 presents the most frequent variant allele, including gene duplication for different populations.[48]

TABLE 31.4 *CYP2D6* Allele Activity Scores, *CYP2D6* Diplotypes, CYP2D6 Metabolic Phenotypes, and Therapeutic Consequences Related to Codeine Therapy

CYP2D6 Allele—Activity Score	*CYP2D6* Diplotype (Star "*" Nomenclature)—Overall Activity Score	Metabolic Phenotype	Codeine Therapeutic Consequence
*1—1 *2—1	*1/*1—2	Normal metabolizer (NM)	Codeine use as provided in package labeling
*4—0 *9—0.5	*4/*9—0.5	Intermediate metabolizer (IM)	Codeine use as provided in package labeling
*5—0 *6xN—0	*5/*6xN—0	Poor metabolizer (PM)	Alternative to codeine indicated due to potential inefficacy
*1—1 *2x3—3	*1/*2x3—4	Ultrarapid metabolizer (UM)	Alternative to codeine indicated due to potential toxicity

TABLE 31.5 Population Frequencies of Example *CYP2D6* Alleles

Population	*CYP2D6* Allele—Activity Score	Frequency (%) in Population
African	*17—0.5	19.98
African American	*17—0.5	18.22
Caucasian	*4—0	18.50
Middle Eastern	*41—0.5	20.37
East Asian	*10—0.5	42.31
South/Central Asian	*10—0.5	19.76
Americas	*4—0	11.28
Oceanian	*1xN—>1	11.83

TABLE 31.6 Example Thiopurine Methyltransferase (TPMT) Gene Diplotypes, Related Phenotypes, and Potential Therapeutic Consequences of Mercaptopurine Use

TPMT Diplotype (Star "*" Nomenclature)	Metabolic Phenotype	Mercaptopurine Therapeutic Consequence
*1/*1	Normal TPMT activity	Mercaptopurine standard initial dose as indicated
*1/*2	Intermediate TPMT activity	Reduced dose as indicated by guidelines
*2/*3A	Low TPMT activity	Extremely reduced dose and decreased frequency of administration as indicated by guidelines

The use of codeine has been decreasing for a number of reasons, including the potential for serious adverse drug reactions, including death.[46] In April of 2017, the USFDA provided a safety announcement stating that codeine and the analgesic tramadol were contraindicated for use in children under the age of 12 years. This population is at increased risk of respiratory depression and death.[49]

TPMT-Mercaptopurine—"Rational drug design" led to the discovery of mercaptopurine (6-MP), with the drug eventually becoming a component of treatment for acute lymphocytic leukemia (ALL).[50] The drug, used as maintenance therapy in combination with methotrexate, represents the foundation drug therapy in childhood ALL.[51] Exposure to 6-MP thioguanine metabolites can result in severe, life-threatening myelosuppression. The exposure to 6-MP is related to a number of variables, including genetics.[52] Thiopurine methyltransferase (TPMT), the protein product of *TPMT*, is responsible for metabolizing 6-MP to form the thioguanine metabolites. In fact, individuals who are homozygous, with two nonfunctional *TPMT* alleles, are expected to experience severe myelosuppression with risk of death.[52] Individuals who carry one no-function *TPMT* allele along with a functional *TPMT* allele have an intermediate risk of myelosuppression.[52] Table 31.6 presents example *TPMT* diplotypes related to myelosuppression risk.

The recognition that the exposure to 6-MP was a pharmacokinetic-based, genetically mediated issue has resulted in a standard of care of preemptive *TPMT* genotyping to minimize the risk of myelosuppression.[52]

The previous examples involve pharmacogenes that are related to drug metabolism with direct dosing/drug selection consequences. Other pharmacogenes of interest, in the pharmacokinetic context, include drug transporters.

Transporter Pharmacogene

SLCO1B1-Simvastatin—Beyond drug-metabolizing gene products (e.g., CYP2C19, CYP2D6, TPMT, etc.), drug transporters as products of PK pharmacogenes also affect drug response.[33] The gene *SLCO1B1* codes for the influx (uptake) transporter organic anion transporter 1B1 (OATP1B1, also called SLCO1B1). OATP1B1, among other transporters, works to move simvastatin and other statins into tissues. Relative to the liver, OATP1B1 moves simvastatin to its site of action where inhibition of HMG CoA reductase results in decreased cholesterol production, thus exerting its pharmacologic and therapeutic effects.[53] Additionally, the liver is the main site of statin metabolism.[54] Decreased transport of simvastatin into hepatocytes has been related to increased risk of myopathy. Variants of *SLCO1B1* impart an increased risk of muscle pain

TABLE 31.7 Example *SLCO1B1* Alleles, Example Diplotypes, Related Phenotypes, and Potential Simvastatin Therapeutic Consequences

SLCO1B1 Allele–Function Status	Example Diplotype	Transporter Phenotype	Simvastatin Therapeutic Consequence
*5—decreased	*1a/*5	Decreased function	Utilize desired dose
*15—decreased	*1b/*15	Decreased function	Utilize a lower dose or consider a different statin
*17—decreased	*5/*17	Poor function	Utilize a lower dose or consider a different statin

(myalgia) and damage, including myopathy and in severe cases rhabdomyolysis.[53] Normal function forms of *SLCO1B1* include *1a and *1b, and the *14 and *35 variants appear to impart increased function. Table 31.7 presents example variant alleles of *SLCO1B1* that have been associated with simvastatin-induced muscle pain and damage.

A combination of a normal function form (e.g., *1a) and an increased function variant (e.g., *14) results in an increased function phenotype where the desired dose can be utilized. Simvastatin exposure in the face of a decreased or poor function phenotype is greater than that for the other statins and the current recommendations for therapeutic considerations apply to this statin with a strong recommendation.[53]

Research has also identified relationships between pharmacodynamic (PD) pharmacogenes and certain drugs. Examples of PD pharmacogenes include those that code for receptors (e.g., beta adrenergic, opiate) and enzymes (nondrug metabolizing; e.g., glucose-6-phosphate dehydrogenase [G6PD] and vitamin K epoxide reductase [VKORC1]). Two specific examples of drug-PD pharmacogene interactions are presented below.

Pharmacodynamic Pharmacogenes

G6PD—Rasburicase—G6PD is necessary to help protect red blood cells from oxidative stress and prevent excessive formation of methemoglobin.[55] Increased oxidative stress can lead to hemolytic anemia, resulting in cardiovascular, renal, liver, and other organ system complications.[55–57] Additionally, red blood cell hemolysis in G6PD deficiency can lead to methemoglobinemia.[58] In this setting, neither oxygen nor carbon dioxide is carried as iron in hemoglobin is in the ferric state rather than the ferrous state.[57,58] Methemoglobinemia may result in death in severe cases.[57–59] There are well over 150 *G6PD* variants and, unlike the previous examples, *G6PD* variants are not designated using the star "*" nomenclature. Individuals with a normal phenotype are not at increased risk of hemolytic anemia, whereas those with a deficient phenotype of any kind are at risk of acute hemolytic anemia. A "variable" phenotype may be identified in some individuals and requires further testing to ascertain risk.[60]

Numerous drugs and chemicals can elicit complications in patients with G6PD deficiency due to a *G6PD* variant or variants.[60] Rasburicase is indicated for use in managing plasma uric acid concentrations in patients with hematologic or solid tumor cancers who are receiving drugs that can cause lysis of cells.[61] Tumor lysis can result in elevated plasma uric acid concentrations, which can cause acute kidney injury.[61] Rasburicase administration in the presence of G6PD variants can result in hemolytic anemia and methemoglobinemia. Rasburicase is contraindicated in persons with a phenotype of "deficient" (of any kind). There are many potential diplotype combinations, which can result in the various phenotype designations. Table 31.8 presents examples of the rasburicase–*G6PD* interaction. Additionally, tables related to drugs to be avoided or monitored in patients who are G6PD deficient are provided in the literature.[60]

TABLE 31.8 Example G6PD Gene Diplotypes, Related Phenotype, and Potential Rasburicase Therapeutic Consequence

Example Diplotype	Phenotype	Rasburicase Therapeutic Consequence
B/Sao Boria	Normal	Utilize drug as indicated
Chatham/Chatham	Deficient	Contraindication to the use of the drug—consider alternative
Villeurbanne/Villeurbanne	Deficient	Contraindication to the use of the drug—consider alternative

TABLE 31.9 *VKORC1* and *CYP2C9* Genotypes and Diplotypes, Related to Predicted Daily Warfarin Dose Requirements

	CYP2C9 Diplotype					
VKORC1	*1/*1	*1/*2	*1/*3	*2/*2	*2/*3	*3/*3
GG	5–7	5–7	3–4	3–4	3–4	0.5–2
AG	5–7	3–4	3–4	3–4	0.5–2	0.5–2
AA	3–4	3–4	0.5–2	0.5–2	0.5–2	0.5–2

VKORC1—Warfarin—There are a number of genes that have been found to affect warfarin dosage requirements. These include variants of *CYP2C9*, and *CYP4F2* and *VKORC1*, the former of which is a PK-related pharmacogene and the latter two of which are PD-related pharmacogenes. The *CYP2C9* variants, *2 and *3 are the most common, result in decreased clearance of warfarin, potentially leading to decreased dose requirements.[62] Here, considering the vitamin K epoxide reductase complex subunit 1 gene (*VKORC1*), individuals with adenine (A) replacing guanine (G) at c.-1639 (c.-1639G>A; rs9923231) are more sensitive to warfarin. According to the product package labeling, individuals who are homozygous for the A allele, regardless of *CYP2C9* diplotype, have a reduced warfarin dose requirement.[63] Dosing algorithms can be applied in this case, along with other information, including genetics to calculate the appropriate dose.[64–66] Table 31.9 shows the impact of both VKORC1 and CYP2C9 on initial warfarin dose (mg/day) recommendations. Algorithms can refine initial dosing and have been recommended for various populations.[64,67]

Numerous other drug–gene interactions have been identified beyond the examples here. Groups such as CPIC and the DPWG continue to develop and publish evidence-based guidelines that provide clinicians with clear ways forward when considering PGx as a component of clinical decision-making. Pharmacists continue to lead the way in the application of PGx across practice settings.

OBJECTIVE 31.3. DESCRIBE PHARMACIST COMPETENCIES IN GENOMICS/PHARMACOGENOMICS AND THEIR ROLES IN PRECISION MEDICINE

Beyond the science of PGx are components related to pharmacist clinical use and application of DNA data. Pharmacist competencies have been developed to address the areas of basic genetic knowledge, genetics and disease, pharmacogenetics and PGx, as addressed above, and ethical, legal, and social implications (ELSI).[68,69] These competencies have been updated from the original set provided by the American Association of Colleges of Pharmacy in 2002.[70] The pharmacist competencies are provided along with other healthcare profession's competencies through the Genetics/Genomics Competency Center (G2C2) of the US National Human Genome Research Institute.[69] In total, these competency statements across the health professions distinguish individual professional expertise.[69] There are four domains of competencies for pharmacists, including the following:

1. Basic genetic concepts;
2. Genetics and disease;
3. Pharmacogenetics/PGx; and
4. ELSI.

Competency domains 1 and 3 were presented earlier to some extent. These competency domains, as well as domains 2 and 4, are addressed briefly here.

Basic Genetic Concepts

Competency domain 1 is related to demonstration of an understanding of basic genetic/genomic concepts and nomenclature. This includes the pharmacist understanding the role of many factors that influence genetics related to the manifestation of disease. The pharmacist must appreciate genetic variations that facilitate strategies related to disease prevention, diagnosis, and treatment. Included in this competency is the expectation that pharmacists realize the differences in PGx testing methodologies. Finally, competency domain 1 relates to the use of the family history over at least three generations in assessing disease risk and selection of drug therapy.[69]

Genetic and Disease

A genetic predisposition to a disease or "genetic susceptibility" is the increased likelihood of developing a specific disease based on an individual's genotype.[71] The genetic basis for many diseases such as cystic fibrosis, hemophilia, sickle-cell anemia, neurofibromatosis, and Huntington's disease has long been established.[72,73] Moreover, the more recent discovery of mutations in the BRCA1 and BRCA2 genes, which are the strongest susceptibility genes for breast and ovarian cancer, has become critical indicators for the prevention, treatment, and prognosis of these cancers in clinical medicine.[74–76] Underlying these advances is the elucidation of the human genome which has led to remarkable progress in determining the genetic basis for more complex, multifactorial diseases.

The patterns of inheritance of genetic diseases can be classified into two broad categories: single gene or monogenic and multifactorial.[77,78] The monogenic or Mendelian pattern which is the simplest pattern of genetic inheritance encompasses autosomal recessive, autosomal dominant, sex-linked, and mitochondrial genes.[72] In autosomal recessive disease inheritance, a trait is expressed in individuals who have two copies (homozygous) of the gene.[72,73,77] Autosomal recessive diseases include cystic fibrosis, alkaptonuria, and phenylketonuyria.[72,77] The autosomal dominant inheritance pattern results in a disease phenotype in individuals who inherit one copy (heterozygous) or two copies (homozygous) of the gene.[72,73,77] Autosomal dominant diseases are Marfan syndrome, a chrondrodysplasia and osteogenesis imperfecta.[72,77] Disease genes can also be transmitted by the X or Y chromosome and are termed sex-linked.[72,73,77] However, an X-linked gene cannot be transmitted from a father to a male son.[72,73,77] Sex-linked diseases include Rett syndrome, Klinefelter syndrome, Duchenne muscular dystrophy, and hemophilia.[72,77] Disease traits can also be transmitted by a cell's mitochondria. The mitochondria, which play a key role in energy metabolism for cells, contain multiple copies of a circular double-stranded DNA. Interestingly, this mitochondrial DNA is only maternally inherited. As a result, mitochondrial disease traits are only passed on to the offspring from the mother.[72,73,77] Mitochondrial diseases include Leigh syndrome, Leber hereditary neuropathy, and fatal infantile neuropathy.[72,77] Though many important diseases are monogenic, the majority of common disease are multifactorial or polygenic.[78–80] Furthermore, multifactorial disease inheritance is inherently more complex as it is phenotypically expressed by a combination of multiple genetic and environment interactions.[78,79] Important multifactorial diseases with known genetic markers include type 1 and 2 diabetes mellitus, hypercholesterolemia, breast, ovarian, and colorectal cancer, familial Alzheimer's disease, schizophrenia, premature coronary heart disease, hypertension, Crohn's disease, and ankylosing spondylitis.[79,81,82]

A critical tool in discovering the genetic markers for these complex, multifactorial diseases is the *genome-wide association study* or GWAS.[82–84] Prior to the development of GWAS, determining specific genetic markers or SNPs for common diseases was limited to searching for the genes predicted to be associated with the disease based on disease's pathophysiology.[72,81,83] This "candidate gene" approach provided some important breakthroughs in finding the specific SNPs for disease but was limited because knowledge of the specific pathophysiology of a disease was not always known.[72,81,83] However, with the technological advances in genetic analysis and the concomitant reduction in the cost of this testing, it has become possible to test an individual's entire genome.[81] Currently, GWAS uses complex SNP arrays to genotype hundreds of thousands of SNPs in large numbers of individual for the multifactorial disease of interest.[81] In addition, with the high throughput analysis now possible with GWAS, genetic markers for a disease can be determined without knowledge of its specific pathophysiology. The use of GWAS has led to advances in the potential genetic underpinnings of an increasing list of multifactorial diseases. However, GWAS has its limitations, including a relative insensitivity to rare variants, the requirement for large sample sizes, the risk of genotyping errors and possible bias in the selection of the individuals for the GWAS.[83] Despite these limitations, however, GWAS will remain a powerful tool to determine the genetic signals that underlie the genetic predisposition to disease.

Pharmacogenetics/Pharmacogenomics

As stated earlier, PGt/PGx for pharmacists relates to understanding the influence of populations relative to polymorphisms and associations of polymorphisms with response to medications. Additionally, pharmacists need to recognize the availability of evidence-based guidelines, such as those from CPIC and the DPWG, which can be applied to patient care.[68,69]

Ethical, Legal, and Social Implications

Perhaps no other competency domain is as common across healthcare professions.[69] This speaks to healthcare professionals, respecting patient privacy and utilizing patient information with confidentiality. Pharmacists as well as other

healthcare professionals understand the need for patient privacy. Although one pharmacist ELSI competency statement points to privacy, emphasis should be placed on security of the DNA data.[68] Here, it is important to understand that genetic data relate to family members, not to the patient only.

Although testing may be intended to provide PGx information, incidental findings may relate to the predisposition of disease. For example, apolipoprotein E (Apo E) variants may be related to certain statin effectiveness in types of familial hypercholesterolemia; however, an APO E variant are also related to the risk of late onset Alzheimer's disease.[69] Clearly, incidental finding of PGx testing must be put in context, and the pharmacist must carefully consider how to utilize this type of data.

Legislation has been enacted at various government levels to help prevent discrimination of individuals relative to healthcare and employment based on genetic information. In the United States, the Genetic Information Nondiscrimination Act was adopted in 2008.[85] State governments have expanded the Federal law to provide further protection of individuals based on their genetics.[86] The previous examples address only a couple of the noted ELSI components of the pharmacist's competency statements in genomics, and the reader is directed to the G2C2 for more detailed information.[69]

Growth and Future Direction of Pharmacogenomics

As PGx testing has previously focused on single drug−gene interactions and has moved to testing multiple genes in panel-based testing, it is envisioned that whole genome sequencing will eventually be a standard of clinical care, moving beyond the research realm of GWAS. This will allow for identification of rare variants that were not tested for in single gene or gene panel tests. Further implementation of PGx will occur and will be placed in a broader context of precision medicine with the potential application of theragnostics, where diagnosis and treatment(s) include molecular biomarkers.[87,88] Drug development is moving to include PGx, and drugs and molecular tests are being marketed together as "companion diagnostic" products.[89]

In specific genetic diseases, applications of biotechnology in genetic medicine, such as gene editing with CRISPR-CAS9, will lead to important treatment advances.[90] In 2012, the European Medicines Agency approved the first gene therapy using an adeno-associated virus vector (Glybera) to treat lipoprotein lipase deficiency, a very rare genetic disease.[91] Similarly, specific gene therapies, such as the first product approved by the US Food and Drug Administration, in 2017, tisagenlecleucel, for childhood and young adult leukemia will provide options that were not available before, giving hope to patients and their families.[92] PGx, however, will have broader application than the above-mentioned advances, and pharmacists can lead the way in PGx implementation.

CONCLUSION

Better understanding of relationships between genes and drug response through technological advancements has resulted in the more precise individualization of therapy. Pharmacogenomic testing is becoming more affordable. PGx, related to PK and PD parameters, are used to improve drug therapy. With defined genomic/pharmacogenomic competencies and teamwork, pharmacists can take the lead in contributing to efficacious and safe treatment being provided to patients early in therapy.